

HOMOLOGICAL DIMENSIONS OF DERIVED HOM COMPLEXES

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ABSTRACT. Let R be a commutative noetherian local ring and M and N be R -complexes with finitely generated homology. We prove that formulas for the homological dimensions of the derived Hom complex $\mathbf{R}\mathrm{Hom}_R(M, N)$, in terms of the homological dimensions of M and N , hold without the *a priori* assumptions on the dimensions of M or N present in classic results.

INTRODUCTION

Throughout this paper, R is a commutative noetherian local ring, and $\mathrm{D}_{\square}^f(R)$ denotes the full subcategory of the derived category, $\mathrm{D}(R)$, whose objects are complexes with bounded and degreewise finitely generated homology. For complexes M and N , in $\mathrm{D}_{\square}^f(R)$ Foxby [8, Thm. 4.1(a)] proved the equality

$$(0.0.1) \quad \mathrm{id}_R \mathbf{R}\mathrm{Hom}_R(M, N) = \mathrm{pd}_R M + \mathrm{id}_R N ;$$

in particular, if two of the quantities $\mathrm{id}_R \mathbf{R}\mathrm{Hom}_R(M, N)$, $\mathrm{pd}_R M$, and $\mathrm{id}_R N$ are finite, then so is the third. Part (b) of the same theorem asserts that the equality

$$(0.0.2) \quad \mathrm{pd}_R \mathbf{R}\mathrm{Hom}_R(M, N) = \mathrm{id}_R M - \mathrm{depth}_R N$$

holds under the assumption that $\mathrm{id}_R N$ is finite. That is, under that assumption, $\mathrm{pd}_R \mathbf{R}\mathrm{Hom}_R(M, N)$ is finite if and only if $\mathrm{id}_R M$ is finite. In this paper we remove the *a priori* assumption on N and prove, in Corollary 3.4, a genuine two-out-of-three statement. That is to say, even though the quantity $\mathrm{id}_R N$ does not show in equality (0.0.2), finiteness of $\mathrm{pd}_R \mathbf{R}\mathrm{Hom}_R(M, N)$ and $\mathrm{id}_R M$ forces it to be finite, and then the equality holds. Similar improvements are made to other results from [8, 9]; they are also found in Section 3. Partial results in this direction—imposing conditions on the homology of $\mathbf{R}\mathrm{Hom}_R(M, N)$ —have recently been obtained by Dey and Ghosh [?], Dey, Holanda, and Miranda-Neto [7], Ghosh and Puthenpurakal [?], Ghosh and Takahashi [?], Holanda, Jorge-Pérez, and Mendoza-Rubio [11], Kimura [13], and by Martins, Mendoza-Rubio, and Nason [14].

Our proofs proceed via stability results for Gorenstein homological dimensions. While it may not be explicitly stated in the literature, one can piece together the equivalence of the following conditions for a complex M in $\mathrm{D}_{\square}^f(R)$:

- (i) $\mathrm{Gpd}_R M$ is finite.
- (ii) $\mathbf{R}\mathrm{Hom}_R(M, R)$ belongs to $\mathrm{D}_{\square}^f(R)$ and $\mathrm{Gpd}_R \mathbf{R}\mathrm{Hom}_R(M, R)$ is finite.

The key step towards our main results is to remove the boundedness assumption on $\mathbf{R}\mathrm{Hom}_R(M, R)$ in part (ii), which is done in Lemma 2.5.

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1. STABILITY OF AUSLANDER AND BASS CATEGORIES REVISITED

Stability properties of Auslander and Bass Categories were studied by Avramov and Foxby [3] and have been revisited by several authors since then. We start by recalling some notation and terminology from [5, 10.3] and, in general, we refer the reader to [5] for any unexplained notation and terminology.

Assume that R has a dualizing complex D and consider the adjunction,

$$\mathbf{D}(R) \xrightleftharpoons[\mathbf{RHom}_R(D, -)]{D \otimes_R^{\mathbf{L}} -} \mathbf{D}(R).$$

For an R -complex M , the unit and counit of this adjunction are the natural morphisms

$$\alpha_D^M : M \longrightarrow \mathbf{RHom}_R(D, D \otimes_R^{\mathbf{L}} M)$$

and

$$\beta_D^M : D \otimes_R^{\mathbf{L}} \mathbf{RHom}_R(D, M) \longrightarrow M.$$

The gross Auslander Category of R is the following full subcategory of $\mathbf{D}(R)$:

$$\hat{\mathbf{A}}(R) = \{M \in \mathbf{D}(R) \mid \text{the unit morphism } \alpha_D^M \text{ is an isomorphism}\}$$

and the Auslander Category of R is

$$\mathbf{A}(R) = \{M \in \hat{\mathbf{A}}(R) \mid M \text{ and } D \otimes_R^{\mathbf{L}} M \text{ have bounded homology}\}.$$

The gross Bass Category of R is the following full subcategory of $\mathbf{D}(R)$:

$$\hat{\mathbf{B}}(R) = \{M \in \mathbf{D}(R) \mid \text{the counit morphism } \beta_D^M \text{ is an isomorphism}\}$$

and the Bass Category of R is

$$\mathbf{B}(R) = \{M \in \hat{\mathbf{B}}(R) \mid M \text{ and } \mathbf{RHom}_R(D, M) \text{ have bounded homology}\}.$$

1.1. Assume that R has a dualizing complex D . For every complex M in $\mathbf{D}^f(R)$ the biduality morphism in $\mathbf{D}(R)$,

$$M \longrightarrow \mathbf{RHom}_R(\mathbf{RHom}_R(M, D), D)$$

is an isomorphism; this is known as Grothendieck Duality. We recall from [5, 18.2.3, 19.1.7, 19.1.12, 19.1.13, 19.4.27] some properties used throughout the paper.

- M belongs to $\mathbf{D}_{\square}^f(R)$ if and only if $\mathbf{RHom}_R(M, D)$ belongs to $\mathbf{D}_{\square}^f(R)$

For a complex M in $\mathbf{D}_{\square}^f(R)$ one further has:

- $\mathrm{pd}_R M$ is finite if and only if $\mathrm{id}_R \mathbf{RHom}_R(M, D)$ is finite.
- $\mathrm{Gpd}_R M$ is finite if and only if $\mathrm{Gid}_R \mathbf{RHom}_R(M, D)$ is finite, equivalently, M belongs to $\mathbf{A}(R)$ if and only if $\mathbf{RHom}_R(M, D)$ belongs to $\mathbf{B}(R)$.

The next theorem is propelled by a result of Avramov, Iyengar, and Lipman [4, Thm. 3.3]; in the proof it is invoked through [5, 19.4.19].

1.2 Theorem. *Assume that R has a dualizing complex.*

- (a) *For a complex M in $\mathbf{D}_{\square}^f(R)$ the following conditions are equivalent.*
- (i) $\mathrm{Gpd}_R M$ is finite.
 - (ii) M belongs to $\mathbf{A}(R)$.
 - (iii) M belongs to $\hat{\mathbf{A}}(R)$.
 - (iv) $\mathbf{RHom}_R(M, D)$ belongs to $\hat{\mathbf{B}}(R)$.

- (b) For a complex N in $D_{\square}^f(R)$ the following conditions are equivalent.
- (i) $\text{Gid}_R N$ is finite.
 - (ii) N belongs to $\mathbf{B}(R)$.
 - (iii) N belongs to $\hat{\mathbf{B}}(R)$.
 - (iv) $\mathbf{RHom}_R(N, D)$ belongs to $\hat{\mathbf{A}}(R)$.

Proof. Let D be a dualizing complex for R .

(a): Conditions (i)–(iii) are equivalent by [5, 19.1.12, 19.4.19]. By [5, 19.1.7] it follows from (ii) that the complex $N = \mathbf{RHom}_R(M, D)$ belongs to $\mathbf{B}(R)$. To prove that (iv) implies (iii) it suffices by Grothendieck Duality to show that the complex $\mathbf{RHom}_R(N, D)$ belongs to $\hat{\mathbf{A}}(R)$. There is a commutative diagram in $D(R)$,

$$\begin{array}{ccc} \mathbf{RHom}_R(N, D) & \xrightarrow[\simeq]{\mathbf{RHom}_R(\beta_D^N, D)} & \mathbf{RHom}_R(D \otimes_R^{\mathbf{L}} \mathbf{RHom}_R(D, N), D) \\ \downarrow \alpha_D^{\mathbf{RHom}_R(N, D)} & & \downarrow \simeq \\ \mathbf{RHom}_R(D, D \otimes_R^{\mathbf{L}} \mathbf{RHom}_R(N, D)) & \xrightarrow{\eta} & \mathbf{RHom}_R(D, \mathbf{RHom}_R(\mathbf{RHom}_R(D, N), D)) \end{array}$$

where $\eta = \mathbf{RHom}_R(D, \eta^{DND})$ is an isomorphism by [5, 12.3.26(b)]. The diagram now shows that $\alpha_D^{\mathbf{RHom}_R(N, D)}$ is an isomorphism, as desired.

(b): Per [5, 19.1.13] the implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) are standard.

(iii) $\Rightarrow$ (iv): Set $M = \mathbf{RHom}_R(N, D)$. If $N \simeq \mathbf{RHom}_R(M, D)$ belongs to $\hat{\mathbf{B}}(R)$ then M belongs to $\hat{\mathbf{A}}(R)$ by part (a).

(iv) $\Rightarrow$ (i): The complex $M = \mathbf{RHom}_R(N, D)$ belongs to $D_{\square}^f(R)$, so it has finite Gorenstein projective dimension by part (a), whence N has finite Gorenstein injective dimension. $\square$

Recall that a complex M in $D_{\square}^f(R)$ is called *perfect* if it is isomorphic in $D(R)$ to a bounded complex of finitely generated free R -modules, equivalently $\text{pd}_R M$ is finite. We recall a few facts about perfect complexes that will be used throughout.

1.3. Let M be a perfect R -complex; one has

$$(1.3.1) \quad -\inf \mathbf{RHom}_R(M, R) = \text{pd}_R M = \text{depth } R - \text{depth}_R M,$$

and $\mathbf{RHom}_R(M, R)$ is itself a perfect complex; see [5, 12.3.20, 19.4.14]. The second equality above is the Auslander–Buchsbaum Formula from [2], see also [5, 16.4.2].

The next result removes the boundedness assumption on M in results of Christensen and Holm [6, Prop. 4.1] and Sather-Wagstaff [15, Prop. 4.1].

1.4 Proposition. Assume that R has a dualizing complex and let M and N be complexes in $D^f(R)$. If N is perfect and not acyclic, then the next assertions hold.

- (a) The following conditions are equivalent.
- (i) M belongs to $\mathbf{A}(R)$.
 - (ii) $N \otimes_R^{\mathbf{L}} M$ belongs to $\mathbf{A}(R)$.
 - (iii) $\mathbf{RHom}_R(N, M)$ belongs to $\mathbf{A}(R)$.
- (b) The following conditions are equivalent.
- (i) M belongs to $\mathbf{B}(R)$.
 - (ii) $N \otimes_R^{\mathbf{L}} M$ belongs to $\mathbf{B}(R)$.
 - (iii) $\mathbf{RHom}_R(N, M)$ belongs to $\mathbf{B}(R)$.

Proof. First recall that if M has bounded homology, then so does $N \otimes_R^{\mathbf{L}} M$; see e.g. [5, 15.4.3]. On the other hand, if $N \otimes_R^{\mathbf{L}} M$ has bounded homology, then M has bounded homology by Iversen's amplitude inequality from [12] as improved by Foxby and Iyengar [10, Thm. 3.1], see also [5, 18.5.10].

Further, there are isomorphisms, see [5, 12.3.26(a)],

$$\begin{aligned} \mathbf{R}\mathrm{Hom}_R(N, M) &\simeq \mathbf{R}\mathrm{Hom}_R(\mathbf{R}\mathrm{Hom}_R(\mathbf{R}\mathrm{Hom}_R(N, R), R), M) \\ &\simeq \mathbf{R}\mathrm{Hom}_R(N, R) \otimes_R^{\mathbf{L}} \mathbf{R}\mathrm{Hom}_R(R, M) \\ &\simeq \mathbf{R}\mathrm{Hom}_R(N, R) \otimes_R^{\mathbf{L}} M. \end{aligned}$$

As $\mathbf{R}\mathrm{Hom}_R(N, R)$ is a perfect complex, M has bounded homology if and only if $\mathbf{R}\mathrm{Hom}_R(N, M)$ has bounded homology.

The assertions now follow from [6, Prop. 4.1] and [15, Prop. 4.1]. $\square$

1.5 Corollary. *Assume that R has a dualizing complex and let M and N be complexes in $\mathbf{D}^f(R)$. If N belongs to $\mathbf{D}_{\square}^f(R)$, is not acyclic, and has finite injective dimension, then the next assertions hold.*

- (a) M belongs to $\mathbf{A}(R)$ if and only if $\mathbf{R}\mathrm{Hom}_R(M, N)$ belongs to $\mathbf{B}(R)$.
- (b) M belongs to $\mathbf{B}(R)$ if and only if $\mathbf{R}\mathrm{Hom}_R(M, N)$ belongs to $\mathbf{A}(R)$.

Proof. Let D be a dualizing complex for R ; by Grothendieck Duality one has

$$\begin{aligned} \mathbf{R}\mathrm{Hom}_R(M, N) &\simeq \mathbf{R}\mathrm{Hom}_R(M, \mathbf{R}\mathrm{Hom}_R(\mathbf{R}\mathrm{Hom}_R(N, D), D)) \\ &\simeq \mathbf{R}\mathrm{Hom}_R(\mathbf{R}\mathrm{Hom}_R(N, D) \otimes_R^{\mathbf{L}} M, D). \end{aligned}$$

The complex $\mathbf{R}\mathrm{Hom}_R(N, D)$ is perfect, so the assertions are now immediate from Proposition 1.4. $\square$

2. DERIVED REFLEXIVE COMPLEXES

A complex M in $\mathbf{D}_{\square}^f(R)$ is called *derived reflexive* if $\mathbf{R}\mathrm{Hom}_R(M, R)$ belongs to $\mathbf{D}_{\square}^f(R)$ and the biduality morphism $M \rightarrow \mathbf{R}\mathrm{Hom}_R(\mathbf{R}\mathrm{Hom}_R(M, R), R)$ is invertible; see [5, 10.2.4]. Every perfect R -complex is derived reflexive, see [5, 19.4.14]. We recall a few facts about these complexes that will be used throughout the paper.

2.1. Let M be a complex in $\mathbf{D}_{\square}^f(R)$; it is derived reflexive if and only if it has finite Gorenstein projective dimension, see [5, 19.4.13], and in that case one has

$$(2.1.1) \quad -\inf \mathbf{R}\mathrm{Hom}_R(M, R) = \mathrm{Gpd}_R M = \mathrm{depth} R - \mathrm{depth}_R M,$$

and $\mathbf{R}\mathrm{Hom}_R(M, R)$ is itself a derived reflexive complex, see [5, 19.4.14]. The second equality above is the Auslander–Bridger Formula from [1], see also [5, 19.4.25]. Finally, if R has a dualizing complex, then M is derived reflexive if and only if it belongs to the Auslander Category $\mathbf{A}(R)$, see [5, 19.1.12].

In several proofs the next remark allows us to assume the existence of a dualizing R -complex.

2.2 Remark. For complexes M and N in $\mathbf{D}_{\square}^f(R)$ the base changed complexes $\widehat{R} \otimes_R M$ and $\widehat{R} \otimes_R N$ belong to $\mathbf{D}_{\square}^f(\widehat{R})$, see [5, 18.3.2], and the complexes

$$\widehat{R} \otimes_R \mathbf{R}\mathrm{Hom}_R(M, N) \simeq \mathbf{R}\mathrm{Hom}_{\widehat{R}}(\widehat{R} \otimes_R M, \widehat{R} \otimes_R N)$$

and

$$\widehat{R} \otimes_R (M \otimes_R^{\mathbf{L}} N) \simeq (\widehat{R} \otimes_R M) \otimes_{\widehat{R}}^{\mathbf{L}} (\widehat{R} \otimes_R N)$$

are in $\mathbf{D}^f(\widehat{R})$; see [5, 12.2.5, 12.2.10, 12.3.29, 12.3.31(a)].

(a) For every complex X in $\mathbf{D}^f(R)$ the projective and Gorenstein projective dimensions of X over R and $\widehat{R} \otimes_R X$ over $\widehat{R}$ are simultaneously finite, see [5, 15.4.21, 17.4.28, 19.3.15, 19.4.3]. In particular, X is a derived reflexive or perfect R -complex if and only if $\widehat{R} \otimes_R X$ is a derived reflexive or perfect $\widehat{R}$ -complex.

(b) For every complex X in $\mathbf{D}^f(R)$ there is an inequality $\mathrm{id}_{\widehat{R}}(\widehat{R} \otimes_R X) \leq \mathrm{id}_R X$, see [5, 13.2.7, 15.4.16(b), 17.3.20]; for X in $\mathbf{D}_{\square}^f(R)$ equality holds, see [5, 18.3.10].

The apparent lack of duality between (0.0.2) and the formula given below for $\mathrm{Gpd}_R \mathbf{RHom}_R(N, M)$ is easily clarified: While the dual invariant to the infimum is the supremum, the infimum of a complex in $\mathbf{D}^f(R)$ agrees per [5, 16.2.5(a)] with the width invariant which is, indeed, dual to the depth.

2.3 Proposition. *Let M and N be complexes in $\mathbf{D}^f(R)$. If N is perfect and not acyclic, then the following conditions are equivalent.*

- (i) M is derived reflexive.
- (ii) $N \otimes_R^{\mathbf{L}} M$ is derived reflexive.
- (iii) $\mathbf{RHom}_R(N, M)$ is derived reflexive.

When these conditions are satisfied, there are equalities,

$$\mathrm{Gpd}_R(N \otimes_R^{\mathbf{L}} M) = \mathrm{pd}_R N + \mathrm{Gpd}_R M$$

and

$$\mathrm{Gpd}_R \mathbf{RHom}_R(N, M) = \mathrm{Gpd}_R M - \inf N .$$

Proof. One can per Remark 2.2 assume that R is complete and, in particular, that it has a dualizing complex. The equivalence of conditions (i)–(iii) is now evident from Proposition 1.4(a).

Assuming that conditions (i)–(iii) hold, one has

$$\begin{aligned} \mathrm{Gpd}_R(N \otimes_R^{\mathbf{L}} M) &= \mathrm{depth} R - \mathrm{depth}_R(N \otimes_R^{\mathbf{L}} M) \\ &= \mathrm{depth} R - (\mathrm{depth}_R N + \mathrm{depth}_R M - \mathrm{depth} R) \\ &= \mathrm{pd}_R N + \mathrm{Gpd}_R M \end{aligned}$$

and

$$\begin{aligned} \mathrm{Gpd}_R \mathbf{RHom}_R(N, M) &= \mathrm{depth} R - \mathrm{depth}_R \mathbf{RHom}_R(N, M) \\ &= \mathrm{depth} R - (\mathrm{depth}_R M + \inf N) \\ &= \mathrm{Gpd}_R M - \inf N . \end{aligned}$$

Here the first, fourth, and sixth equalities holds by the Auslander–Bridger Formula (2.1.1). The second equality holds by [5, 16.3.1], the third holds by the Auslander–Bridger and Auslander–Buchsbaum Formulas (2.1.1) and (1.3.1). Finally, the fifth equality holds by [5, 16.2.25]. $\square$

2.4 Proposition. *Assume that R has a dualizing complex and let M and N be complexes in $D^f(R)$. If N is perfect, then the following conditions are equivalent.*

- (i) $\text{Gid}_R M$ is finite and M belongs to $D_\square^f(R)$.
- (ii) $\text{Gid}_R(N \otimes_R^{\mathbf{L}} M)$ is finite and $N \otimes_R^{\mathbf{L}} M$ belongs to $D_\square^f(R)$.
- (iii) $\text{Gid}_R \mathbf{R}\text{Hom}_R(N, M)$ is finite.

When these conditions are satisfied, there are equalities,

$$\text{Gid}_R(N \otimes_R^{\mathbf{L}} M) = \text{Gid}_R M - \inf N$$

and

$$\text{Gid}_R \mathbf{R}\text{Hom}_R(N, M) = \text{Gid}_R M + \text{pd}_R N.$$

Proof. The equivalence of the three conditions is a consequence of [5, 19.1.13] and Proposition 1.4(b).

Assuming that conditions (i)–(iii) hold, one has

$$\begin{aligned} \text{Gid}_R(N \otimes_R^{\mathbf{L}} M) &= \text{depth}_R R - \inf(N \otimes_R^{\mathbf{L}} M) \\ &= \text{depth}_R R - (\inf N + \inf M) \\ &= \text{Gid}_R M - \inf N \end{aligned}$$

and

$$\begin{aligned} \text{Gid}_R \mathbf{R}\text{Hom}_R(N, M) &= \text{depth}_R R - \inf \mathbf{R}\text{Hom}_R(N, M) \\ &= \text{depth}_R R - (\inf M + \text{depth}_R N - \text{depth}_R R) \\ &= \text{depth}_R R - \inf M + (\text{depth}_R R - \text{depth}_R N) \\ &= \text{Gid}_R M + \text{pd}_R N. \end{aligned}$$

The first, third, and fourth equalities hold by the Bass Formula for Gorenstein injective dimension, see [5, 19.2.40], the second is standard, see [5, 16.2.10]. The fifth equality holds by [5, 16.3.5(b)], and the last equality holds by the Auslander–Buchsbaum Formula (1.3.1) and another application of [5, 19.2.40]. $\square$

The equivalence of (i) and (ii) in the next lemma is known from [4, Thm. 2.1]. The difference between conditions (ii) and (iii) is that the latter does not require the homology of the complex $\mathbf{R}\text{Hom}_R(M, R)$ to be bounded below.

2.5 Lemma. *Let M be a complex in $D_\square^f(R)$; the next conditions are equivalent.*

- (i) M is derived reflexive.
- (ii) $\mathbf{R}\text{Hom}_R(M, R)$ is derived reflexive.
- (iii) $\text{Gpd}_R \mathbf{R}\text{Hom}_R(M, R)$ is finite.

Proof. Per 2.1 the implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) are known. To finish the proof, assume that $\text{Gpd}_R \mathbf{R}\text{Hom}_R(M, R)$ is finite. Per Remark 2.2 one can assume that R has a dualizing complex; call it D . By [5, 19.1.12] the complex $\mathbf{R}\text{Hom}_R(M, R)$ belongs to $\hat{\mathbf{A}}(R)$. By [5, 19.1.2] the isomorphisms

$$\mathbf{R}\text{Hom}_R(M, R) \simeq \mathbf{R}\text{Hom}_R(M, \mathbf{R}\text{Hom}_R(D, D)) \simeq \mathbf{R}\text{Hom}_R(D, \mathbf{R}\text{Hom}_R(M, D))$$

now show that $\mathbf{R}\text{Hom}_R(M, D)$ belongs to $\hat{\mathbf{B}}(R)$, whence M is derived reflexive by Theorem 1.2(a). $\square$

2.6 Theorem. *Let M and N be complexes in $D_{\square}^f(R)$. If N is perfect and not acyclic, then the following conditions are equivalent.*

- (i) M is derived reflexive.
- (ii) $\mathbf{R}\mathrm{Hom}_R(M, N)$ is derived reflexive.
- (iii) $\mathrm{Gpd}_R \mathbf{R}\mathrm{Hom}_R(M, N)$ is finite.

When these conditions are satisfied, one has

$$\mathrm{Gpd}_R \mathbf{R}\mathrm{Hom}_R(M, N) = \mathrm{pd}_R N - \inf M .$$

Proof. The implication (i) $\Rightarrow$ (ii) follows from Lemma 2.5 by induction on the number of non-zero modules in the complex of free R -modules isomorphic to N in $D(R)$. Per Remark 2.2 it is also a special case of [6, Cor. 5.6]. The implication (ii) $\Rightarrow$ (iii) is standard. To see that (iii) implies (i), assume $\mathrm{Gpd}_R \mathbf{R}\mathrm{Hom}_R(M, N)$ is finite. Since the following isomorphisms hold,

$$\begin{aligned} \mathbf{R}\mathrm{Hom}_R(M, N) &\simeq \mathbf{R}\mathrm{Hom}_R(M, \mathbf{R}\mathrm{Hom}_R(\mathbf{R}\mathrm{Hom}_R(N, R), R)) \\ &\simeq \mathbf{R}\mathrm{Hom}_R(\mathbf{R}\mathrm{Hom}_R(N, R) \otimes_R^{\mathbf{L}} M, R) , \end{aligned}$$

and the complex $\mathbf{R}\mathrm{Hom}_R(N, R)$ is perfect, the complex M is derived reflexive by Lemma 2.5 and Proposition 2.3.

Assuming that conditions (i)–(iii) hold, one has

$$\begin{aligned} \mathrm{Gpd}_R \mathbf{R}\mathrm{Hom}_R(M, N) &= \mathrm{depth}_R R - \mathrm{depth}_R \mathbf{R}\mathrm{Hom}_R(M, N) \\ &= \mathrm{depth}_R R - (\mathrm{depth}_R N + \inf M) \\ &= \mathrm{pd}_R N - \inf M . \end{aligned}$$

The first equality is the Auslander–Bridger Formula (2.1.1), the second holds by [5, 16.2.25], and the third is the Auslander–Buchsbaum Formula (1.3.1). $\square$

2.7 Corollary. *Assume that R has a dualizing complex and let M and N be complexes in $D_{\square}^f(R)$. If M is not acyclic and has finite injective dimension, then the following conditions are equivalent.*

- (i) $\mathrm{Gid}_R N$ is finite
- (ii) $\mathbf{R}\mathrm{Hom}_R(M, N)$ is derived reflexive.
- (iii) $\mathrm{Gpd}_R \mathbf{R}\mathrm{Hom}_R(M, N)$ is finite.

When these conditions are satisfied, one has

$$\mathrm{Gpd}_R \mathbf{R}\mathrm{Hom}_R(M, N) = \mathrm{id}_R M - \mathrm{depth}_R N .$$

Proof. The implication (ii) $\Rightarrow$ (iii) is standard. Now let D be a dualizing complex for R and note that by Grothendieck Duality, there are isomorphisms,

$$\begin{aligned} \mathbf{R}\mathrm{Hom}_R(M, N) &\simeq \mathbf{R}\mathrm{Hom}_R(M, \mathbf{R}\mathrm{Hom}_R(\mathbf{R}\mathrm{Hom}_R(N, D), D)) \\ &\simeq \mathbf{R}\mathrm{Hom}_R(\mathbf{R}\mathrm{Hom}_R(N, D), \mathbf{R}\mathrm{Hom}_R(M, D)) . \end{aligned}$$

The complex $\mathbf{R}\mathrm{Hom}_R(M, D)$ is perfect. Thus, if (i) holds, then $\mathbf{R}\mathrm{Hom}_R(N, D)$ is derived reflexive, whence $\mathbf{R}\mathrm{Hom}_R(M, N)$ is derived reflexive by Theorem 2.6. If (iii) holds, then the complex $\mathbf{R}\mathrm{Hom}_R(N, D)$ is derived reflexive by Theorem 2.6, and so $\mathrm{Gid}_R N$ is finite.

Assuming that conditions (i)–(iii) hold one has

$$\begin{aligned} \mathrm{Gpd}_R \mathbf{R}\mathrm{Hom}_R(M, N) &= \mathrm{depth}_R R - \mathrm{depth}_R \mathbf{R}\mathrm{Hom}_R(M, N) \\ &= \mathrm{depth}_R R - (\mathrm{depth}_R N + \inf M) \\ &= \mathrm{id}_R M - \mathrm{depth}_R N. \end{aligned}$$

Here the first equality is the Auslander–Bridger Formula (2.1.1), the second equality holds by [5, 16.2.25] and the third by the Bass Formula, see [5, 16.4.11]. $\square$

3. PERFECT COMPLEXES

The equivalence of conditions (i) and (ii) in the next lemma is known from [4, Thm. 4.1]. As in Lemma 2.5, the difference between conditions (ii) and (iii) is that the latter does not require the homology of $\mathbf{R}\mathrm{Hom}_R(M, R)$ to be bounded below.

3.1 Lemma. *Let M be a complex in $D_{\square}^f(R)$; the next conditions are equivalent.*

- (i) M is perfect.
- (ii) $\mathbf{R}\mathrm{Hom}_R(M, R)$ is perfect.
- (iii) $\mathrm{pd}_R \mathbf{R}\mathrm{Hom}_R(M, R)$ is finite.

Proof. It is clear that the implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) hold. To finish the proof, assume that $\mathrm{pd}_R \mathbf{R}\mathrm{Hom}_R(M, R)$ is finite. By Lemma 2.5 the complex M is derived reflexive. In particular, the complex $\mathbf{R}\mathrm{Hom}_R(M, R)$ belongs to $D_{\square}^f(R)$, so it is perfect and hence so is $M \simeq \mathbf{R}\mathrm{Hom}_R(\mathbf{R}\mathrm{Hom}_R(M, R), R)$. $\square$

The next result improves [5, 16.4.19], which we know from Foxby’s notes [9].

3.2 Theorem. *Let M and N be non-acyclic complexes in $D_{\square}^f(R)$. If two of the conditions (1)–(3) below are satisfied, then so is the third condition, and $\mathbf{R}\mathrm{Hom}_R(M, N)$ is a perfect complex with*

$$\mathrm{pd}_R \mathbf{R}\mathrm{Hom}_R(M, N) = \mathrm{pd}_R N - \inf M.$$

- (1) M is perfect.
- (2) N is perfect.
- (3) $\mathrm{pd}_R \mathbf{R}\mathrm{Hom}_R(M, N)$ is finite.

Proof. If M is perfect, then $\mathbf{R}\mathrm{Hom}_R(M, N)$ belongs to $D_{\square}^f(R)$ by [5, 15.4.3], and [5, 16.4.19] yields $\mathrm{pd}_R \mathbf{R}\mathrm{Hom}_R(M, N) = \mathrm{pd}_R N - \inf M$. Thus, it suffices to show that (1) follows from (2) and (3). Since the following isomorphisms hold

$$\begin{aligned} \mathbf{R}\mathrm{Hom}_R(M, N) &\simeq \mathbf{R}\mathrm{Hom}_R(M, \mathbf{R}\mathrm{Hom}_R(\mathbf{R}\mathrm{Hom}_R(N, R), R)) \\ &\simeq \mathbf{R}\mathrm{Hom}_R(\mathbf{R}\mathrm{Hom}_R(N, R) \otimes_R^{\mathbf{L}} M, R), \end{aligned}$$

the complex $\mathbf{R}\mathrm{Hom}_R(N, R) \otimes_R^{\mathbf{L}} M$ is perfect by Lemma 3.1, whence M is perfect by [8, Thm. 4.2(a)], which is also [5, 16.4.17]. $\square$

The next result improves [8, Thm. 4.1(b)] which is also [5, 16.4.34].

3.3 Corollary. *Let M and N be non-acyclic complexes in $D_{\square}^f(R)$. If two of the conditions (1)–(3) below are satisfied, then so is the third condition, and $\mathbf{R}\mathrm{Hom}_R(M, N)$ is a perfect complex with*

$$\mathrm{pd}_R \mathbf{R}\mathrm{Hom}_R(M, N) = \mathrm{id}_R M - \mathrm{depth}_R N.$$

- (1) $\mathrm{id}_R M$ is finite.
- (2) $\mathrm{id}_R N$ is finite.
- (3) $\mathrm{pd}_R \mathbf{R}\mathrm{Hom}_R(M, N)$ is finite.

Proof. If $\text{id}_R N$ is finite, then $\mathbf{R}\text{Hom}_R(M, N)$ belongs to $\mathbf{D}_{\square}^f(R)$ by [5, 15.4.9], and [5, 16.4.34] yields $\text{pd}_R \mathbf{R}\text{Hom}_R(M, N) = \text{id}_R M - \text{depth}_R N$. Thus, it suffices to show that (2) follows from (1) and (3). Per Remark 2.2 one can assume that R has a dualizing complex, call it D . The following isomorphisms hold

$$\begin{aligned} \mathbf{R}\text{Hom}_R(M, N) &\simeq \mathbf{R}\text{Hom}_R(M, \mathbf{R}\text{Hom}_R(\mathbf{R}\text{Hom}_R(N, D), D)) \\ &\simeq \mathbf{R}\text{Hom}_R(\mathbf{R}\text{Hom}_R(N, D), \mathbf{R}\text{Hom}_R(M, D)). \end{aligned}$$

The complex $\mathbf{R}\text{Hom}_R(M, D)$ is perfect, so $\mathbf{R}\text{Hom}_R(N, D)$ is perfect by Theorem 3.2, whence $\text{id}_R N$ is finite. $\square$

For complexes M and N in $\mathbf{D}_{\square}^f(R)$ it is shown in [8, Thm. 4.2(a)] that

$$\text{pd}_R(M \otimes_R^{\mathbf{L}} N) = \text{pd}_R M + \text{pd}_R N$$

holds without further assumptions, while $\text{id}_R(M \otimes_R^{\mathbf{L}} N) = \text{id}_R M - \inf N$ by part (b) of the same theorem holds under the assumption that N is perfect. The next result improves [8, Thm. 4.2(b)], which is also [5, 16.4.35].

3.4 Theorem. *Let M and N be non-acyclic complexes in $\mathbf{D}_{\square}^f(R)$. If two of the conditions (1)–(3) below are satisfied, then so is the third condition, and $M \otimes_R^{\mathbf{L}} N$ is a complex in $\mathbf{D}_{\square}^f(R)$ of finite injective dimension:*

$$\text{id}_R(M \otimes_R^{\mathbf{L}} N) = \text{id}_R M - \inf N.$$

- (1) $\text{id}_R M$ is finite.
- (2) N is perfect.
- (3) $\text{id}_R(M \otimes_R^{\mathbf{L}} N)$ is finite.

Proof. If N is perfect, then $M \otimes_R^{\mathbf{L}} N$ belongs to $\mathbf{D}_{\square}^f(R)$, see [5, 15.4.3], and [5, 16.4.35] yields $\text{id}_R(M \otimes_R^{\mathbf{L}} N) = \text{id}_R M - \inf N$. Thus, it suffices to show that (2) follows from (1) and (3). Set $\widehat{M} = \widehat{R} \otimes_R M$ and $\widehat{N} = \widehat{R} \otimes_R N$; by Remark 2.2 the $\widehat{R}$ -complexes $\widehat{M}$ and $\widehat{M} \otimes_{\widehat{R}}^{\mathbf{L}} \widehat{N}$ have finite injective dimension, and it suffices to show that $\widehat{N}$ is a perfect $\widehat{R}$ -complex.

Let D be a dualizing complex for $\widehat{R}$. It suffices to show that the complex $\widehat{M} \otimes_{\widehat{R}}^{\mathbf{L}} \widehat{N}$ has bounded homology. Indeed, it then belongs to $\mathbf{D}_{\square}^f(\widehat{R})$, so the dual complex

$$\mathbf{R}\text{Hom}_{\widehat{R}}(\widehat{M} \otimes_{\widehat{R}}^{\mathbf{L}} \widehat{N}, D) \simeq \mathbf{R}\text{Hom}_{\widehat{R}}(\widehat{N}, \mathbf{R}\text{Hom}_{\widehat{R}}(\widehat{M}, D))$$

is perfect, and since $\mathbf{R}\text{Hom}_{\widehat{R}}(\widehat{M}, D)$ is perfect, so is $\widehat{N}$ by Theorem 3.2.

As $\widehat{M}$ belongs to the Bass category, see [5, 10.3.5], there are isomorphisms,

$$\begin{aligned} \widehat{M} \otimes_{\widehat{R}}^{\mathbf{L}} \widehat{N} &\simeq (D \otimes_{\widehat{R}}^{\mathbf{L}} \mathbf{R}\text{Hom}_{\widehat{R}}(D, \widehat{M})) \otimes_{\widehat{R}}^{\mathbf{L}} \widehat{N} \\ &\simeq D \otimes_{\widehat{R}}^{\mathbf{L}} (\mathbf{R}\text{Hom}_{\widehat{R}}(D, \widehat{M}) \otimes_{\widehat{R}}^{\mathbf{L}} \widehat{N}). \end{aligned}$$

As $\mathbf{R}\text{Hom}_{\widehat{R}}(D, \widehat{M})$ is perfect, the complex $\mathbf{R}\text{Hom}_{\widehat{R}}(D, \widehat{M}) \otimes_{\widehat{R}}^{\mathbf{L}} \widehat{N}$ belongs to $\mathbf{D}_{\square}^f(R)$, see [5, 15.4.9, 15.4.3]. Since $\widehat{M} \otimes_{\widehat{R}}^{\mathbf{L}} \widehat{N}$ belongs to $\widehat{\mathbf{B}}(R)$, again by [5, 10.3.5], the complex $\mathbf{R}\text{Hom}_{\widehat{R}}(D, \widehat{M}) \otimes_{\widehat{R}}^{\mathbf{L}} \widehat{N}$ belongs to $\widehat{\mathbf{A}}(R)$, see [5, 19.1.2], and therefore to $\mathbf{A}(R)$ by Theorem 1.2(a). Now another application of [5, 19.1.2] shows that $\widehat{M} \otimes_{\widehat{R}}^{\mathbf{L}} \widehat{N}$ belongs to $\mathbf{B}(R)$; in particular, it has bounded homology. $\square$

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